

Combining Deterministic Rules and Lightweight LLMs for Explainable Oncology Trial Matching

Abstract—Screening patients for oncology clinical trials is slow and error-prone, as it requires manually reconciling free-text clinical records with dense, protocol-specific eligibility criteria. Purely LLM-driven matching pipelines improve throughput but offer limited transparency, since eligibility decisions emerge from opaque probabilistic reasoning rather than auditable logic. We propose a clinician-in-the-loop pipeline that separates information extraction from eligibility reasoning: unstructured oncology notes and trial protocols are converted into structured patient profiles and logical rules, respectively, and matching is resolved deterministically whenever a rule maps onto the structured schema, falling back to LLM-assisted reasoning only for criteria that cannot be formalized. Across three open-source 8B-parameter models tested on synthetic oncology diaries and real trial protocols, the pipeline reached 92.3% accuracy on eligibility-criteria extraction, 78.3% on patient-profile extraction, and 84.0% on the final matching task, with clinician-rated justification quality of 4.93/5. The results indicate that lightweight, rule-grounded LLM pipelines can deliver both competitive accuracy and the traceability required for clinical adoption.

Index Terms—Large Language Models; Clinical Trial Matching; Information Extraction; Rule-Based Reasoning; Human-in-the-Loop; Oncology

INTRODUCTION

Identifying eligible patients for oncology clinical trials remains a challenging and time-consuming task, as clinicians must manually cross-reference narrative clinical records against lengthy and often complex eligibility criteria. LLMs are increasingly used to automate parts of this process, but most published pipelines either assume structured EHR input or resolve eligibility end-to-end through free-form generation, which makes individual decisions difficult to audit and reproduce.

We argue that reliable clinical decision support requires decoupling *what the patient’s record says* from *how eligibility is decided*. This paper introduces a three-stage pipeline that (i) extracts structured patient profiles from free-text oncology notes, (ii) formalizes trial eligibility criteria into machine-checkable logical expressions, and (iii) matches patients to trials by evaluating each criterion deterministically whenever possible, escalating to LLM reasoning only for criteria that resist formalization. Every intermediate artifact is reviewed by a clinician before being used downstream.

Using three 8B open-source LLMs over synthetic oncology notes and real ClinicalTrials.gov protocols, the pipeline achieves up to 92.3% accuracy on criteria extraction, 78.3% on patient-profile extraction, and 84.0% on patient-trial matching, with expert-rated justification scores averaging 4.93/5. These

results show that small, openly available models, combined with deterministic reasoning where possible, can offer a practical and auditable alternative to end-to-end probabilistic matching.

RELATED WORK

LLMs have proven effective at pulling structured entities such as diagnoses, biomarkers, and treatments out of clinical narratives [1], [3], [9], [11], [12], but this line of work generally stops at extraction and does not connect its output to a downstream decision workflow [9], [11].

For trial matching specifically, prevailing approaches lean on semantic retrieval or fully generative reasoning to decide eligibility [4], [8], [12]. These systems can perform well but rarely expose *why* a given patient was matched or excluded [4], [12], and many still presuppose structured EHR access that is uncommon in routine oncology practice [4], [6].

Our framework differs by treating extraction and eligibility reasoning as distinct, clinician-audited stages, and by defaulting to deterministic rule evaluation rather than free-form generation whenever a criterion can be mapped onto structured data. This extends hybrid designs explored in prior work [3], [4] toward a pipeline where LLM involvement is minimized and localized precisely to the cases that require it, improving traceability across the full matching process.

PROPOSED FRAMEWORK

The framework targets patient-screening workload reduction while keeping a clinician in control of every artifact the system produces. It is organized into three stages: (1) patient parameter extraction, (2) criteria extraction and formalization, and (3) semi-deterministic matching.

Patient Parameter Extraction

Free-text oncology notes selected by a clinician are cleaned of irrelevant content and passed to the LLM with a schema-specific prompt. The model populates a predefined patient schema and normalizes diagnoses against a stakeholder-agreed mapping. The clinician reviews and, if needed, corrects the resulting profile before it is stored for later use (Fig. 1).

Criteria Extraction and Conversion

Trial protocols are pre-processed to isolate eligibility-relevant sections. The LLM first detects the trial’s cohort

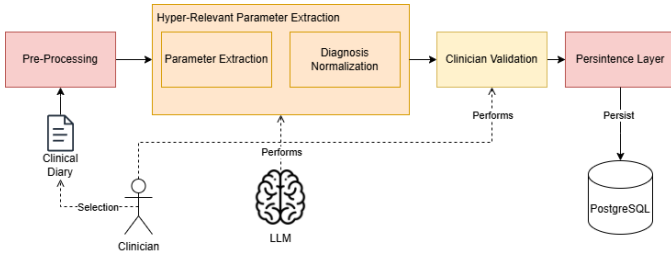


Fig. 1. Patient parameter extraction workflow.

structure, then extracts inclusion/exclusion criteria, keeping each criterion linked to its cohort and splitting compound statements only when this does not distort their clinical meaning. Extracted criteria and their subsequent translation into structured logical expressions are each independently validated by a clinician before being released for matching (Fig. 2).

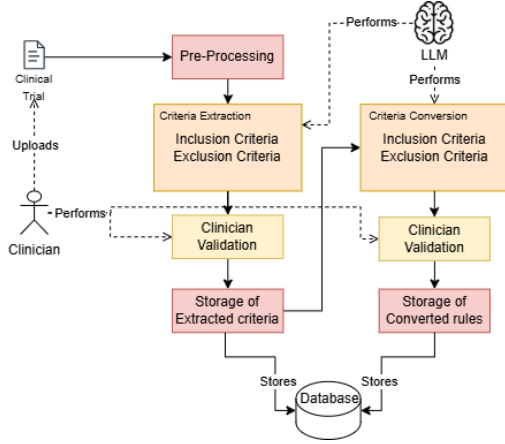


Fig. 2. Criteria extraction and formalization workflow

Semi-Deterministic Matching

For a selected trial, candidate patients are pre-filtered by pathology group to limit unnecessary comparisons. Each remaining patient is checked against every eligibility criterion using one of two strategies: a *deterministic* check when the criterion maps onto a structured schema field, or an *LLM-assisted* check, using the profile, clinical diary, and lab results, when it does not, in which case the model also returns a justification for expert review. Both strategies yield a uniform criterion-level verdict. The full set of verdicts is presented to the clinician for confirmation or correction, and every decision, override, and justification is logged for auditability (Fig. 3).

EXPERIMENTAL EVALUATION

Datasets and Models

Evaluation used two datasets: 30 synthetic oncology clinical diaries generated with `gpt-oss-120b` and grounded in four real diaries to preserve realistic style and noise, and 30 real trial protocols from *ClinicalTrials.gov* spanning varied tumour

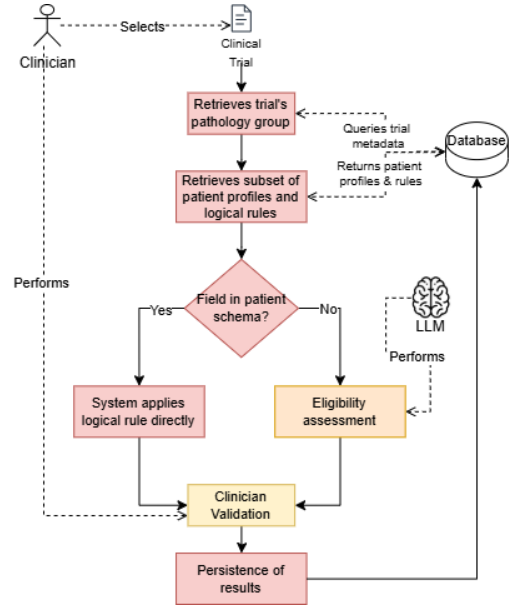


Fig. 3. Semi-deterministic patient-trial matching workflow.

types, stages, molecular profiles, and treatment settings. Three open-source 8B models, Qwen3 8B Instruct, Llama 3.1 8B Instruct, and DeepSeek-R1 8B, were tested with deterministic decoding and zero-shot prompting, also, no fine-tuning was applied.

Results

Table I reports the best model per task. All tasks but the final matching step used a three-way scheme (correct/partial/wrong), from which we report both strict accuracy (fully correct only) and partial-aware accuracy (correct + partial).

TABLE I
BEST-MODEL PERFORMANCE PER PIPELINE TASK.

Task	Best Model	Accuracy
Patient Profile Extraction	Qwen3	78.3%/83.3%
Trial Structure Identification	DeepSeek	96.7%/96.7%
Criteria Extraction	Qwen3	92.3%/95.9%
Criteria Extraction (cohort-level)	Qwen3	96.3%/96.3%
Criteria Conversion	DeepSeek	64.1%/73.4%
Patient Matching	Qwen3	84.0%/84.0%

Patient profile extraction. All models produced schema-valid JSON 100% of the time. Qwen3 led with 78.3% strict / 83.3% partial accuracy and the lowest missing-field rate (11.6%); DeepSeek followed closely (74.3%/80.5%, 13.3% missing), while Llama3.1 trailed (62.3%/70.3%, 21.7% missing). Simple fields (Gender, ECOG-PS) were solved perfectly by all models; diagnosis normalization, hierarchical treatment history, TNM staging, and dates accounted for most errors, reflecting difficulty with temporal and multi-level structure rather than entity recognition itself (Fig. 4).

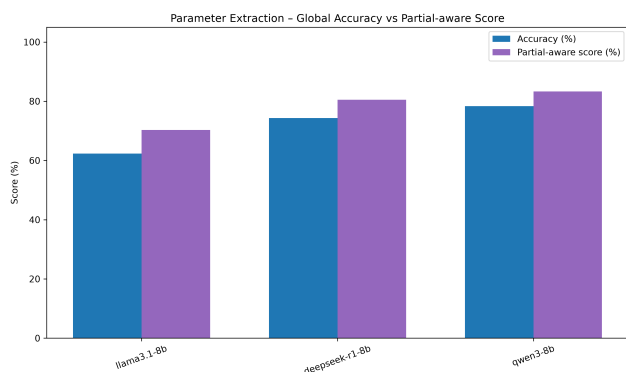


Fig. 4. Patient parameter extraction: strict vs. partial accuracy.

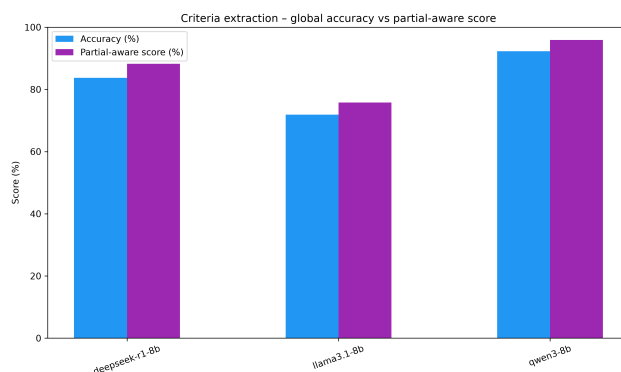


Fig. 6. Criteria extraction: strict vs. partial accuracy.

Trial structure identification. Results varied sharply by model: DeepSeek reached 96.7% strict accuracy, Qwen3 60% (75% partial), and Llama3.1 only 25%, mainly confusing descriptive patient groups with formal cohorts and struggling with schema compliance. Expert review noted that correctly identifying a cohort did not guarantee its clinical meaning was preserved; most errors involved omitted eligibility-defining details rather than wrong cohort boundaries (Fig. 5).

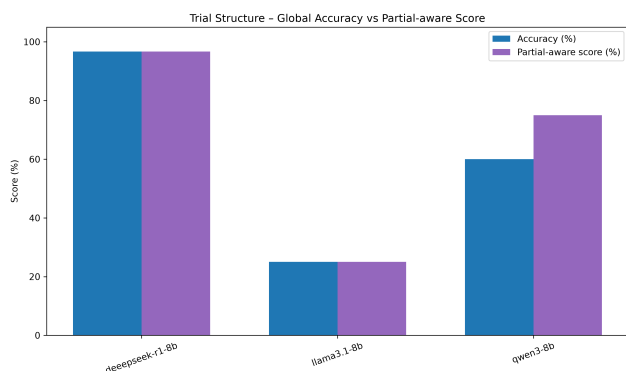


Fig. 5. Trial structure identification: strict vs. partial accuracy.

Criteria extraction. This was the strongest stage overall: Qwen3 reached 92.3%/95.9% (strict/partial) on individual criteria and 96.3% on cohort attribution; DeepSeek followed with 83.7%/88.2% and 66.7%; Llama3.1 trailed at 71.9%/75.8% and 66.7%, generating the most incorrect criteria. Expert review confirmed Qwen3 best preserved clinical meaning even when restructuring compound statements, DeepSeek avoided hallucinations, and Llama3.1 more often fragmented compound criteria in ways that changed their logic, errors that were mostly representational rather than interpretive, underscoring the value of clinician validation at this step (Figs. 6–7).

Criteria conversion. Formalizing criteria into logical expressions was the pipeline’s bottleneck. Qwen3 led with 65.1%/74.4% but also produced the most outright errors; Llama3.1 was comparable (62.5%/73.5%); DeepSeek had the lowest strict score (52.2%) but a substantial increase in partial

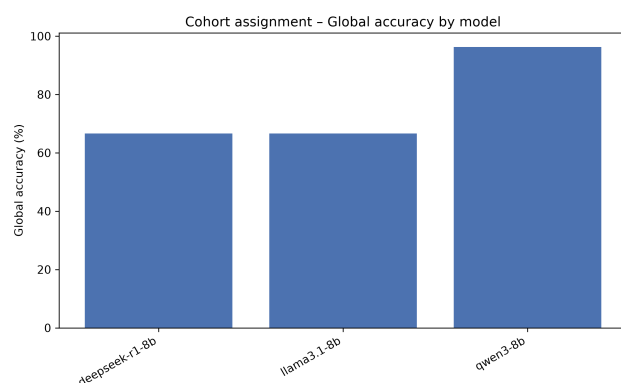


Fig. 7. Cohort attribution accuracy.

score (71.2%). The dominant failure mode across models was under-representing compound criteria with nested AND/OR logic, typically capturing only the primary condition. A secondary issue was misassigning criteria to non-schema fields when a matching schema field already existed, especially for treatment history. Testing a larger model (gpt-oss-120b) improved robustness, 65.1%/80.9% accuracy with far fewer outright errors (35), though its residual errors were mostly schema-normalization mismatches rather than logical omissions (Fig. 8).

Patient-trial matching. Despite weaker conversion scores, final matching recovered strongly: Qwen3 84.0%, DeepSeek 83.0%, Llama3.1 78.0%. Clinician-rated justification quality followed the same ranking, Qwen3 4.93/5, grounding decisions in explicit evidence and defaulting to caution on missing data; DeepSeek 4.8/5, reliably distinguishing “absent” from “unavailable” information; Llama3.1 4.2/5, occasionally over-interpreting uncertainty (e.g., rejecting patients despite adequate lab values). This confirms that reliability at this stage depends as much on transparent justification as on raw accuracy (Fig. 9).

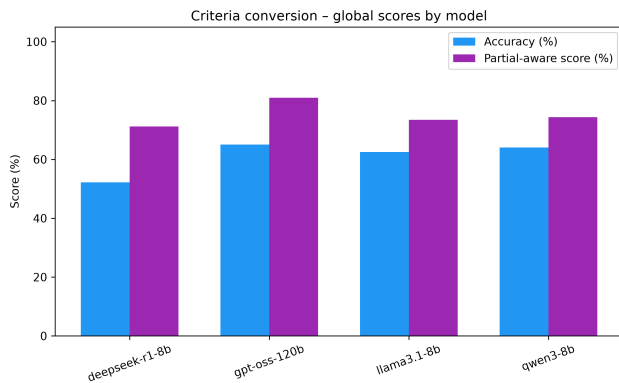


Fig. 8. Criteria conversion: strict vs. partial accuracy.

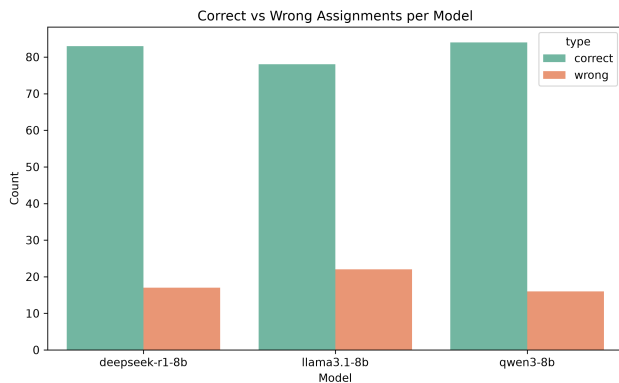


Fig. 9. Patient-trial matching: correct vs. wrong.

DISCUSSION

Structured extraction consistently outperformed logical formalization across all three models, confirming that translating free-text criteria into machine-checkable logic, not entity recognition, is the pipeline’s main obstacle. No single model dominated every task: Qwen3 was strongest at extraction, while DeepSeek excelled at trial-structure identification and matching reliability, suggesting that decomposing the workflow lets each stage draw on whichever model handles it best, rather than forcing a single end-to-end solution.

More broadly, the results support routing eligibility decisions through deterministic logic whenever possible and reserving LLM reasoning for the residual, non-formalizable criteria. Combined with mandatory clinician review at each stage, this design curbs reliance on opaque probabilistic outputs and yields the auditability that clinical decision-support tools require.

CONCLUSION

We presented a clinician-validated pipeline that pairs LLM-based extraction with deterministic rule evaluation for oncology trial matching, escalating to LLM reasoning only where formal rules cannot capture a criterion. Lightweight

8B models proved capable across the pipeline, with Qwen3 reaching 78.3% on patient-profile extraction, 92.3% on criteria extraction, and 84.0% on final matching; logical formalization remained the weakest link and is a priority for future work. Overall, the results suggest that structured representations, deterministic reasoning, and clinician oversight together offer a practical, explainable path toward automated oncology trial recruitment. Future work will test larger models, strengthen criteria formalization, and validate the pipeline on larger real-world clinical corpora.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work, the author(s) used Large Language Models, such as ChatGPT, or similar tools for the refining of the work’s contents. The author(s) reviewed and edited the output as needed and take full responsibility for the content of the published article.

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